

Practical – 2

Problem Statement –

Extend the portfolio application from Practical 1 by adding React Router. Implement a navigation bar with at least 3 routes: Home, Projects, and Contact. Each route must render a distinct component without a full page reload. Add at least 2 useState variables used meaningfully one for toggling UI visibility and one for managing a form input on the Contact page. The Contact page must include a controlled input that captures and displays user input in real time.

Code –

/src/main.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import { BrowserRouter } from 'react-router-dom';
import App from './App.jsx';
import './App.css';

createRoot(document.getElementById('root')).render(
  <StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </StrictMode>,
)
```

/src/components/Navbar.jsx

```
import { useState } from 'react';
import { NavLink } from 'react-router-dom';

function Navbar() {
  // useState #1: toggles mobile nav menu visibility
  const [menuOpen, setMenuOpen] = useState(false);

  const toggleMenu = () => setMenuOpen((prev) => !prev);

  const navLinks = [
    { path: '/', label: 'cd ~' },
    { path: '/projects', label: 'cd ~/projects' },
    { path: '/contact', label: 'cd ~/contact' },
  ];

  return (
    <nav className="navbar">
      <div className="navbar-top">
        <span className="navbar-prompt">student@portfolio:~$</span>
        <button
          className="menu-toggle"
          onClick={toggleMenu}
          aria-label="Toggle navigation menu"
        >
          {menuOpen ? '[ close ]' : '[ menu ]'}
        </button>
      </div>
    </nav>
  )
}
```

```

<ul className={`nav-links ${menuOpen ? 'nav-links-open' : ''}`}>
  {navLinks.map((link) => (
    <li key={link.path}>
      <NavLink
        to={link.path}
        end={link.path === '/' }
        className={({ isActive }) => (isActive ? 'nav-link active' : 'nav-link')}
        onClick={() => setMenuOpen(false)}
      >
        {link.label}
      </NavLink>
    </li>
  ))}
</ul>
</nav>
);
}

export default Navbar;

```

/src/pages/Projects.jsx

```

function Projects() {
  const projects = [
    {
      name: "credit-fraud-detection",
      description:
        "Hybrid anomaly detection system combining Isolation Forest, One-Class SVM, LOF, and an Autoencoder with supervised stacking, plus ablation analysis.",
      tech: ["Python", "Scikit-learn", "PyTorch"],
    },
    {
      name: "student-burnout-eda",
      description:
        "EDA on a 37-feature survey dataset examining how AI tool usage correlates with burnout among CS/IT students.",
      tech: ["Pandas", "Matplotlib", "Seaborn"],
    },
    {
      name: "house-price-prediction",
      description:
        "Regression models (Random Forest, Gradient Boosting) on the King County housing dataset with log-transformed targets.",
      tech: ["Scikit-learn", "NumPy"],
    },
    {
      name: "veritas-rag-engine",
      description:
        "RAG-based hallucination-reduction engine using recursive cross-verification and a citation-attribution module.",
      tech: ["Python", "LangChain", "Vector DB"],
    },
  ];
}

```

```

return (
  <section className="page-section projects-page">
    <h2>student@portfolio:~$ ls -la projects/</h2>
    <div className="projects-grid">
      {projects.map((project) => (
        <article className="project-card" key={project.name}>
          <h3>{project.name}</h3>
          <p>{project.description}</p>
          <div className="project-tech">
            {project.tech.map((t) => (
              <span key={t} className="tech-tag">{t}</span>
            ))}
          </div>
        </article>
      ))}
    </div>
  </section>
);
}

export default Projects;

```

/src/pages/Contact.jsx

```

import { useState } from 'react';

function Contact() {
  // useState #2: manages controlled form input, updated in real time
  const [formData, setFormData] = useState({
    name: "",
    email: "",
    message: "",
  });

  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData((prev) => ({ ...prev, [name]: value }));
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    alert(`Message captured:\n\nName: ${formData.name}\nEmail: ${formData.email}\nMessage:
    ${formData.message}`);
  };

  return (
    <section className="page-section contact-page">
      <h2>student@portfolio:~$ ./send_message.sh</h2>

      <form className="contact-form" onSubmit={handleSubmit}>
        <label htmlFor="name">name:</label>
        <input
          type="text"
          id="name"
          name="name"

```

```

      value={formData.name}
      onChange={handleChange}
      placeholder="Enter your name"
      required
    />

    <label htmlFor="email">email:</label>
    <input
      type="email"
      id="email"
      name="email"
      value={formData.email}
      onChange={handleChange}
      placeholder="Enter your email"
      required
    />

    <label htmlFor="message">message:</label>
    <textarea
      id="message"
      name="message"
      value={formData.message}
      onChange={handleChange}
      placeholder="Type your message..."
      rows={4}
      required
    />

    <button type="submit">[ submit ]</button>
  </form>

  { /* Controlled input value displayed live as the user types */ }
  <div className="live-preview">
    <p className="preview-label">student@portfolio:~$ cat draft.txt</p>
    <pre className="preview-output">
      {`name: ${formData.name || '_'}
email: ${formData.email || '_'}
message: ${formData.message || '_'}`}
    </pre>
  </div>
</section>
);
}

export default Contact;

```

/src/App.jsx

```

import { Routes, Route } from 'react-router-dom';
import Navbar from './components/Navbar';
import Footer from './components/Footer';
import Home from './pages/Home';
import Projects from './pages/Projects';
import Contact from './pages/Contact';
import './App.css';

```

```
function App() {
  const studentName = "Taniksha Shah";

  return (
    <div className="page">
      <div className="terminal-window">
        <div className="terminal-titlebar">
          <div className="terminal-buttons">
            <span className="btn btn-red"></span>
            <span className="btn btn-yellow"></span>
            <span className="btn btn-green"></span>
          </div>
          <div className="terminal-title">student@portfolio: ~</div>
        </div>

        <div className="terminal-body">
          <Navbar />

          <main className="main-content">
            <Routes>
              <Route path="/" element={ <Home /> } />
              <Route path="/projects" element={ <Projects /> } />
              <Route path="/contact" element={ <Contact /> } />
            </Routes>
          </main>

          <Footer name={studentName} />
        </div>
      </div>
    </div>
  );
}

export default App;
```

Screenshot –

```

student@portfolio:~$
cd - cd ~/projects cd ~/contact

$ whoami

student@portfolio:~$ whoami
Taniksha Shah

# Aspiring Machine Learning Engineer

student@portfolio:~$ cat about.txt
About Me
I'm a Computer Engineering student specializing in Data Science and
Machine Learning. I build end-to-end ML systems - from data
pipelines and model architecture to deployment - with a focus on
turning research ideas into working, reliable products.

// tech_stack
Python      PyTorch    Scikit-Learn  Pandas & NumPy
SQL         React      Git & GitHub   Data
                                   Visualization

student@portfolio:~$ 0 1000 functions Read / 4433 with Read = 1034

```

```

student@portfolio:~$
cd - cd ~/projects cd ~/contact

student@portfolio:~$ ls -l projects/

credit-fraud-
detection/
  Fraud anomaly
  detection system
  combining Isolation
  Forest, Bag-Classifier,
  SVM, LRF, and an
  Autoencoder with
  supervised stacking,
  plus detection
  analysis.
  Python
  Scikit-Learn
  PyTorch

student-harvest-
yield/
  FFA on a 37-Feature
  survey dataset
  examining how 42
  fuel usage
  correlates with
  harvest using CHRT
  outputs.
  Pandas, Matplotlib
  Seaborn

house-price-
prediction/
  Regression model
  (Random Forest,
  Gradient Boosting)
  on the Big Query
  housing dataset with
  log-transformed
  targets.
  Scikit-Learn
  NumPy

various-rag-
engine/
  RAG-based
  retrieval-augmented
  generation engine
  using recursive
  cross-verification

```

